



QA/4HANA Studio
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PRODUCT DOCUMENTATION · CURRENT VERIFIED BUILD

How-to Guide and Frequently Asked Questions

Task recipes, operating guidance, safety rules, troubleshooting and concise answers for everyday QA/4HANA Studio use.

DOCUMENT Q4H-HOWTO-FAQ-2.0	PRODUCT BASELINE QA/4HANA Studio 2.1.0 development line; v2.0.0 remains frozen
PREPARED 30 July 2026	CLASSIFICATION Internal product and training documentation

Screens containing configured target or account information have been anonymized. Passwords and credential values are never included.

QUICK REFERENCE

Golden path

1. Configure and verify a non-production SAP target.
2. Capture reusable controls in Control Object Repository.
3. Create and publish typed Tests in Compose.
4. Create and validate transaction data in Test Data.
5. Build sequential Business Processes and independent Regression Packs.
6. Preflight, review and run from Execution Center.
7. Review the one canonical evidence document in Audit & Evidence.

Golden safety rules. Do not expose credentials; do not bypass preflight; do not auto-reverse failed business transactions; do not overwrite prior run evidence; and do not run transactional Tests without approved target, data ownership and owner reference.

TASK-ORIENTED GUIDANCE

1 How-to recipes

HOW-01

Configure the SAP connection safely

1. Open the profile menu and select Settings.
2. Choose SAP under Test-system connections.
3. Enter the approved non-production URL, username and password.
4. Save the connection; confirm only non-secret status is shown afterward.
5. Classify and verify the target before live scan or execution.

HOW-02

Create a reusable Single Test

1. Open Compose and search the Test Library first.
2. Select New Test; enter business name, file name, application and process area.
3. Choose Blank or copy an existing Test as a template.
4. Declare typed inputs/outputs and sensitivity.
5. Add steps, objects and value sources; save as Draft.
6. Resolve publish findings and publish.

HOW-03

Model one header with many line items

1. Choose nested JSON when the children naturally belong inside each transaction.
2. Create one header object and add all owned line items to its collection.
3. Preview child count and mappings.
4. Save and use preflight to confirm one transaction with N child items.

HOW-04**Model multiple headers with multiple line items**

1. Create one transaction object per header, each with its own child collection; or use relational CSV.
2. For relational CSV, select unique header key and matching child foreign key.
3. Validate that there are no duplicate headers or orphan children.
4. Preview transaction and child totals before saving.

HOW-05**Create a sequential Business Process**

1. Open Processes & Packs → Business Processes.
2. Create or open a Draft Process.
3. Add published Tests in business order.
4. Bind each stage input from process data, literal, system context or a prior output.
5. Resolve topology/contract validation, then publish.

HOW-06**Create an independent Regression Pack**

1. Open Processes & Packs → Regression Packs.
2. Create a Draft Pack and add Test or Business Process members.
3. Set member-specific data/application/session/failure policies.
4. Validate references and compatible policies.
5. Publish and select it later through Saved Pack execution.

HOW-07**Run a controlled Test or scenario**

1. Open Execution Center and choose the execution type.
2. Select exact scope and App ID.
3. Choose data, filters, limits and headed/headless behavior.
4. Run preflight and correct all blockers.
5. Review immutable data/scope, acknowledge warnings and start.
6. Monitor the stable run route and open canonical evidence when complete.

HOW-08**Rerun safely**

1. Open the original run and choose rerun.
2. Choose full or eligible failed scope.
3. Review input, artifact, target and policy differences.
4. For transactional work, confirm retained state is eligible; started transactions may be blocked.
5. Start the new run and preserve the parent/child lineage.

HOW-09**Find evidence**

1. Open Audit & Evidence.
2. Search by process/Test, App ID, run ID or executor.
3. Filter outcome, mode and date.
4. Open the stable run detail and select Evidence PDF.
5. Confirm the Test name header, AI ELK logo, run identity, target context and redaction.

SAFE RECOVERY

2 Troubleshooting decision guide

Symptom	Check first	Safe action
Studio could not start; HTML was parsed as JSON	Server/API route and built client asset are from the same build.	Verify the API endpoint directly, rebuild, restart the Studio server and reload without changing workspace data.
SAP connection cannot be saved	Credential-vault availability and encrypted fallback state.	Use the server-supported fallback, save again and confirm only non-secret status is returned.
Live scan shows network access denied	VPN/proxy/firewall and server host reachability.	Restore the approved network path; do not work around it by embedding credentials or disabling browser security.
Publish is blocked	Typed contract, required values, objects, parameters and output collisions.	Open each actionable finding, correct the owning artifact and rerun validation.
Preflight is blocked	Target freshness, credentials, versions, data bindings and transactional safeguards.	Follow the exact correction link; rerun preflight to obtain a new plan hash.
Run fails mid-transaction	Failure location, retained SAP state and evidence.	Stop downstream work, retain the run, review evidence and request authorization for any remediation.
Rerun is not eligible	Whether the transaction already started and retains business state.	Do not force it. Create an approved remediation plan with new data/owner reference if required.
Evidence appears different between workspaces	Canonical evidence URL and run ID.	Use the Audit & Evidence canonical PDF; report any route mismatch as a defect.

34 PRODUCT ANSWERS

3 Frequently asked questions

1. What is the difference between a Test, Business Process and Regression Pack?

A Test is one reusable executable automation asset. A Business Process sequences Tests and allows typed outputs to feed later stages. A Regression Pack groups independent Tests or Processes; members do not hand data to each other.

2. What happened to Chain, Suite and Batch?

They remain as compatibility concepts. The current business language is Single Test, Business Process, Pack · Tests, Pack · Processes and Saved Pack. Legacy plans translate into the versioned execution model.

3. When should I use a Business Process?

Use it when order matters or a later Test requires data produced by an earlier Test—for example PO → Goods Receipt → Supplier Invoice.

4. When should I use a Regression Pack?

Use it for independent coverage that can be reported together without cross-member data hand-offs, such as smoke, release or application-area regression collections.

5. What is a Saved Pack?

It is a persisted, versioned Regression Pack created in Processes & Packs. A published mixed Pack can be selected and executed directly in Execution Center.

6. Why is preflight mandatory?

Preflight is the server-owned safety gate. It validates target, credentials, artifacts, versions, contracts, objects, mappings, data, policies and calculated work before SAP is opened.

7. What does the target badge mean?

It shows the configured application, sanitized target identity, safety classification and verification state. A stale or unverified target can block execution.

8. Where are SAP credentials stored?

The server uses the OS credential vault where available and an encrypted local fallback for detached Windows sessions. Passwords are never returned to the browser.

9. Can I use Google sign-in without losing existing data?

Yes. The first verified Google account becomes the preserved single workspace owner. Linking the owner does not move, rename or filter existing artifacts, history or repository data.

10. Are multiple users and roles supported?

No. Version 2.0 intentionally uses one preserved owner identity. Multi-user roles are deferred until the requirement and authorization model are approved.

11. What is the manual duration multiplier?

It is a modeling assumption that estimates how much longer a human execution may take relative to automation runtime. It is not an observed fact or confidence interval and should be reviewed with manual-minutes-per-Test and cost assumptions.

12. Why can potential cost saved be negative?

The model subtracts automation runtime, development, maintenance, licensing, infrastructure and review costs from estimated manual-equivalent cost. A negative number means the modeled investment has not yet been recovered.

13. What data formats are supported?

The Studio supports flat CSV, nested JSON transactions and relational CSV header/child models. Preflight resolves the exact effective records into an immutable snapshot.

14. How do I represent one sales order with many line items?

Use one nested transaction with an items collection, or a unique header row joined to multiple child rows by a foreign key. Do not duplicate the header for each item unless the source contract explicitly expects that format.

15. How do I represent multiple sales orders with line items?

Use multiple transaction objects, each owning its items, or multiple unique header rows joined to their respective child rows. Each header becomes an isolated execution iteration.

16. What are Test inputs and outputs?

Inputs are typed values required by a Test. Outputs are typed values produced for evidence or downstream stages. Sensitivity identifies business or secret-bearing values so authoring and redaction policies can act appropriately.

17. What is a prior-output binding?

It maps an output from an earlier Business Process stage to a required input of a later stage. Forward references, cycles, unknown outputs and incompatible types are rejected.

18. Why can I save a Draft but not publish it?

Drafts allow incomplete authoring. Publishing requires reviewed contracts and valid parameters, objects, value sources, outputs and policies. The editor shows blocking findings that must be corrected.

19. What happens when a Test fails?

The current transaction stops and later stages do not run. Earlier values and evidence are retained for review. The product does not automatically reverse or erase the business transaction.

20. Why is there no automatic reversal?

Automatic reversal could destroy the compliance trail or perform an unapproved business action. Remediation is handled as a separately authorized run with its own immutable evidence.

21. How does progress work for one Business Process?

Known stages, steps and child items contribute determinate progress. Persisted top-level progress is available after reconnect; the complete six-level reconciled event hierarchy remains an open backlog item.

22. What is the difference between headed and headless execution?

Headed opens a visible browser for observation or training. Headless runs without a visible window. Both use the same controlled plan and evidence model; some live diagnostics are easier to observe in headed mode.

23. Can a rerun overwrite the failed run?

No. A rerun creates a new immutable snapshot and records parent/child lineage. Difference review shows changes in data, artifacts, target and policy before execution.

24. What is failed-scope rerun?

It creates a new run containing only eligible failed transactions or Pack members. Transactional items that already started and retain business state may be blocked for safety.

25. Why is there only one evidence PDF?

Execution Center and Audit & Evidence resolve the same canonical evidence URL. This prevents conflicting document versions.

26. What does evidence contain?

It includes branded Test/scenario identity, run ID, outcome, timing, steps, screenshots or observations, target/executor context and source references, subject to redaction and retention policy.

27. How do I find a previous run?

Use Audit & Evidence search for process/Test name, App ID, run ID or executor, then filter outcome, mode and date. Server-side pagination and full lineage traversal are planned enhancements.

28. How do I change the colour theme?

Open the profile menu, choose Settings and select the appearance option there.

29. What should I do if “Scan a live screen” cannot reach SAP?

Confirm the configured URL, target verification, network/VPN path and session health. A network-access-denied message is an environment connectivity issue, not a cue to expose or re-enter credentials in the scan field.

30. What should I do if saved SAP credentials appear unavailable in a detached Windows session?

Save again after the encrypted fallback is available, confirm non-secret credential status, and restart only the affected Studio server session if required. Never paste the password into logs or support tickets.

31. What does “Legacy ready” mean?

The existing Test remains executable through inferred legacy metadata, but publishing under the typed model requires a reviewed contract and current validation.

32. How are secrets kept out of logs and snapshots?

Execution plans and data snapshots reject credential-shaped fields, the runtime redacts known and generic authorization values, and the repository runs a high-confidence secret scan.

33. Are scheduled executions and notifications available?

No. Scheduling, notifications and controlled parallelism are deliberately deferred until target capacity, isolation, cancellation and queue governance are designed.

34. Where can I see product test quality?

Use the QA/4HANA Test Operations companion app. Its catalogue is generated from repository tests and recorded runs and retains historical failures even after remediation.

PRODUCT LANGUAGE

4 Glossary

Term	Meaning
Test	One reusable executable automation asset with steps and a typed contract.
Business Process	An ordered sequence of Tests with explicit typed data hand-offs.
Regression Pack	A versioned collection of independent Test and/or Business Process members.
Iteration / transaction	One isolated effective data record and its owned children.
Preflight	Server-owned validation and impact calculation performed before SAP is opened.
Effective data	The exact filtered, limited and relationally resolved records that will execute.
Immutable snapshot	Frozen plan, bindings and transaction data identified by a hash.
Evidence	The canonical retained execution document and supporting artifacts.
Lineage	Persistent relationship between an original run and a new rerun/remediation run.
Fail-stop	Policy that stops later stages after failure while retaining earlier state and evidence.